

Clinical Trials: An Overview

1. Introduction

Clinical trials are research studies conducted in human participants to evaluate the safety, efficacy, and potential side effects of new drugs, biological products, medical devices, or treatment regimens. These trials are essential to ensure that a new intervention is safe and effective before it is approved for general use.

2. Objectives of Clinical Trials

- To determine the safety of a new intervention.
- To evaluate the efficacy or effectiveness.
- To identify side effects and risks.
- To compare the new intervention with standard treatments.
- To gather information that will support regulatory approval.

3. Phases of Clinical Trials

- Phase I (Safety Study): Conducted on a small number of healthy volunteers (20-100). Focus is on safety, dosage range, and side effects.
- Phase II (Efficacy Study): Conducted on a larger group of patients (100-300). Focus is on efficacy and side effects.
- Phase III (Confirmatory Study): Conducted on large groups (300-3000+). Focus is on confirming efficacy, monitoring side effects, and comparing with standard treatment.
- Phase IV (Post-marketing Study): Conducted after approval to gather information on long-term effects and overall benefits/risks.

4. Types of Clinical Trials

- Interventional Trials - Participants receive the intervention.
- Observational Trials - Researchers observe outcomes without intervention.
- Randomized Controlled Trials (RCT) - Participants are randomly assigned to intervention or control groups.
- Non-Randomized Trials - Assignment is not random.